28. Zhao, F., Wu, B., Wang, L., Sang, H.: A large language model-assisted reinforcement learning framework with evolutionary algorithm for hybrid flow-shop scheduling. IEEE Transactions on Evolutionary Computation (2026)

This appendix provides supplementary material for Section 5.

– Appendix A reports robustness results under alternative variation probabilities and supports the empirical findings in Section 5.1.
– Appendix B presents a representative GPT-5.4-generated selection operator and supports the pattern analysis in Section 5.2.

A Robustness to Different Variation Probabilities

As a robustness check, Tables 6 and 7 report the training- and test-set results from a supplementary experiment with crossover probability 0.8 and mutation probability 0.2. The results are consistent with the main experiment. On training, Claude Sonnet 4.6 and Gemini 3.1 Pro remain the strongest models in the aggregate summaries, while Kimi K2.5 still yields the strongest single training operator according to the best-score column. On test, Claude Sonnet 4.6 and Gemini 3.1 Pro again remain among the top-performing models, with only modest shifts in rankings across the summary metrics.

Table 6: Zero-shot **training** performance summary across benchmarked LLMs with $N = 200$, $G = 100$, crossover probability 0.8, and mutation probability 0.2. Parenthetical integers rank models within each metric column.

Model	Mean over mean	Median over mean	Mean over median	Median over median	Best score	Mean rank
Claude Sonnet 4.6	**0.7025 (1)**	**0.6946 (1)**	**0.7972 (1)**	**0.7946 (1)**	0.8418 (5)	**1.80**
Gemini 3.1 Pro	0.6981 (2)	0.6930 (2)	0.7939 (2)	0.7893 (2)	0.8504 (4)	2.40
GLM-5	0.6889 (3)	0.6893 (3)	0.7711 (3)	0.7763 (3)	0.8510 (3)	3.00
Kimi K2.5	0.6868 (4)	0.6889 (4)	0.7598 (4)	0.7670 (4)	**0.8776 (1)**	3.40
GPT-5.4	0.6646 (6)	0.6793 (5)	0.7111 (5)	0.7418 (6)	0.8557 (2)	4.80
Xiaomi mimo-v2-flash	0.6657 (5)	0.6739 (6)	0.6993 (6)	0.7519 (5)	0.8009 (6)	5.60
Gemini 3.1 Flash-Lite	0.6539 (7)	0.6670 (7)	0.6970 (7)	0.7382 (7)	0.7892 (7)	7.00
DeepSeek-v3.2	0.6211 (8)	0.6271 (8)	0.6156 (8)	0.5753 (8)	0.7664 (8)	8.00

Table 7: Zero-shot **test** performance summary across benchmarked LLMs with $N = 200$, $G = 100$, crossover probability 0.8, and mutation probability 0.2. Parenthetical integers rank models within each metric column.

Model	Mean over mean	Median over mean	Mean over median	Median over median	Best score	Mean rank
Claude Sonnet 4.6	0.5439 (2)	**0.5609 (1)**	**0.5938 (1)**	0.5832 (2)	0.6419 (3)	**1.80**
Gemini 3.1 Pro	**0.5535 (1)**	0.5514 (2)	0.5899 (2)	0.5823 (3)	**0.6454 (1)**	**1.80**
Kimi K2.5	0.5306 (5)	0.5402 (6)	0.5836 (3)	**0.5889 (1)**	0.6440 (2)	3.40
GLM-5	0.5439 (2)	0.5471 (3)	0.5803 (4)	0.5784 (4)	0.6017 (6)	3.80
Gemini 3.1 Flash-Lite	0.5350 (4)	0.5468 (4)	0.5634 (5)	0.5680 (7)	0.6108 (5)	5.00
GPT-5.4	0.5280 (6)	0.5355 (7)	0.5590 (6)	0.5770 (6)	0.6174 (4)	5.80
Xiaomi mimo-v2-flash	0.5227 (7)	0.5439 (5)	0.5488 (7)	0.5781 (5)	0.5975 (7)	6.20
DeepSeek-v3.2	0.4678 (8)	0.5057 (8)	0.5100 (8)	0.5262 (8)	0.5899 (8)	8.00

Algorithm 2 A representative GPT-5.4-generated selection operator

1: **Require:** population $\mathcal{P} = \{p_i\}_{i=1}^n$, number of parents $k \in \mathbb{N}$, stage $t \in [0,1]$, stability constant $\varepsilon > 0$; prediction matrix $P \in \mathbb{R}^{n \times d}$; per-individual mean training errors $\bar{e} = (\bar{e}_1, \dots, \bar{e}_n)^\top$; syntax-tree depths $h = (h_1, \dots, h_n)^\top$ with $h_i = \mathrm{height}(p_i)$; program sizes $|p| = (|p_1|, \dots, |p_n|)^\top$, e.g. node counts.

2: $S \leftarrow \mathrm{clip}(PP^\top, [-1, 1])$ ▷ *clipped similarity matrix*

3: $d \leftarrow \mathbf{1}_n - \frac{1}{n} S \mathbf{1}_n$ ▷ *prediction-space dispersion*

4: Min–max normalize within $\mathcal{P}$; for $i = 1, \dots, n$,

$$q_i^{\mathrm{fit}} \leftarrow 1 - \frac{\bar{e}_i - \min_\ell \bar{e}_\ell}{\max_\ell \bar{e}_\ell - \min_\ell \bar{e}_\ell + \varepsilon}, \quad q_i^{\mathrm{div}} \leftarrow \frac{d_i - \min_\ell d_\ell}{\max_\ell d_\ell - \min_\ell d_\ell + \varepsilon}$$

$$q_i^{\mathrm{sz}} \leftarrow 1 - \frac{|p|_i - \min_\ell |p|_\ell}{\max_\ell |p|_\ell - \min_\ell |p|_\ell + \varepsilon}, \quad q_i^{\mathrm{ht}} \leftarrow 1 - \frac{h_i - \min_\ell h_\ell}{\max_\ell h_\ell - \min_\ell h_\ell + \varepsilon}$$

5: $\nu \leftarrow 0.65(1 - t) + 0.15$ ▷ *diversity blend weight*

6: $\beta \leftarrow 0.15 + 0.55t$ ▷ *parsimony blend weight*

7: Composite score: fit, diversity, and parsimony, weighted by evolutionary stage.

$$q \leftarrow (1.2 + 1.8t)\, q^{\mathrm{fit}} + \nu\, q^{\mathrm{div}} + \tfrac{\beta}{2}(q^{\mathrm{sz}} + q^{\mathrm{ht}})$$

8: $m_e \leftarrow \min\big(n, \lceil k(0.08 + 0.12t) \rceil\big)$ ▷ *number of elite slots*

9: $\mathcal{E} \leftarrow \big\{\text{top } m_e \text{ indices by } q\big\}$ ▷ *elite parent indices*

10: $\theta \leftarrow \max(0.05,\, 0.9 - 0.7t)$ ▷ *softmax temperature*

11: $\ell \leftarrow q/\theta - \max(q)/\theta$ ▷ *shifted selection logits*

12: $\delta \leftarrow 0.92 - 0.25(1 - t)$ ▷ *similarity crowding threshold*

13: Crowding-adjusted weights; softmax mass for selection.

$$g \leftarrow \big(\mathbf{1} + (S > \delta)\mathbf{1}_n\big)^{-1}$$

$m \leftarrow \exp(\ell) \odot (0.35 + 0.65g)$ ▷ *unnormalized softmax weights*

14: For each elite index $i \in \mathcal{E}$, down-weight m_i.

15: $\omega \leftarrow m / \sum_j m_j$ ▷ *softmax selection probabilities*

16: Sample non-elite parents i.i.d. from $\mathrm{Cat}(\omega)$; merge set with elites into $\mathcal{S}$.

17: **return** $\mathcal{S}$ with $|\mathcal{S}| = k$.

B Representative GPT-5.4-Generated Selection Operator

In Algorithm 2, the key idea of the GPT-5.4-designed operator is phenotypic diversity control. Rather than distinguishing individuals only by their mean training error $\bar{e}_i$, it uses their semantics, represented by output vectors in the prediction matrix P, to identify phenotypic duplicates and de-emphasize them during parent selection. Concretely, the operator steers the mating pool toward behaviorally diverse individuals early in the run when stage t is small, while still allowing stronger emphasis on fitness in later generations.

This behavior is realized through the similarity matrix $S = PP^\top$ and the derived dispersion score d_i. In Algorithm 2, Line 7 combines fitness, diversity, and parsimony into a stage-dependent score, with the diversity weight ν in Line 5 decreasing as t grows so that search gradually shifts from exploration toward exploitation. Line 8 preserves a small elite block of high-scoring parents, while Line 13 down-weights individuals in high-similarity neighborhoods before Boltzmann selection. This hybrid design keeps strong parents but avoids filling the mating pool with semantically similar programs.